Quantum sensing for water-quality changes

Classical simulation of proposed quantum sensing | 7 October 2026

27.92% lower modeled total RMSE in an assumed 16-tap pilot

1.67× lower optical error at equal launched probe power

A dissolved-salt proxy calibrated from published optical data

Simulated feasibility; no claim of a deployed chemical detector

The monitoring problem and our contribution

Evidence for the need, a testable optical idea

UN 2025: only 60,000 of ~2 million reporting measurements came from poorest regions.

EPA: online monitoring supports treatment decisions and contamination investigation.

Proposed role: flag a mean salt-proxy change for confirmatory sampling.

Our addition: combine published CV sensing with water calibration, loss and drift tests.

A measurable water proxy

NaCl concentration changes are calibrated; pollutant identity remains unresolved

At 1550 nm: $n(W) = -0.0800\,W^3 + 0.0740\,W^2 + 0.1620\,W + 1.3162$

W is salt mass fraction. Around 500 mg/kg, $dn/dc = 1.621 \times 10^{-7}$ RIU per mg/kg.

Proposed 10 μm interaction length gives 6.570×10^{-6} rad per mg/kg.

YSI EXO field reference: salinity accuracy ±1% or 0.1 ppt; temperature accuracy ±0.01 °C.

Field accuracy bounds are not shot-noise standard deviations. This is a change detector, not a safety test.

Implementation and measurement model

Python on a CPU models a future squeezed-light instrument

NumPy models the Gaussian state; SciPy computes probabilities and constants.

pandas reads and writes CSV data; Matplotlib produces ten scientific figures.

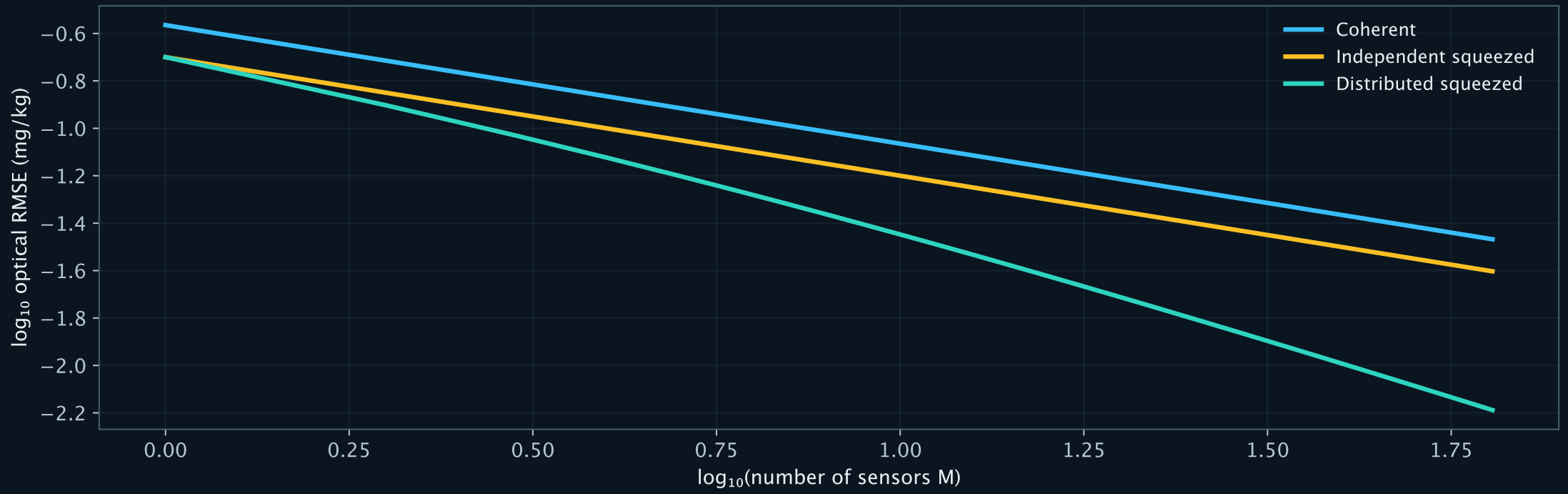
Input: configuration, tap geometry and synthetic local concentration changes.

Calculate optical covariance, $\hat{q} = \Sigma\, y_i / M$, then optical and technical RMSE.

Compare coherent, independent squeezed and distributed probes at matched photons.

Ideal scaling emerges from the model

Lossless and uncapped: squeezed photons increase with M

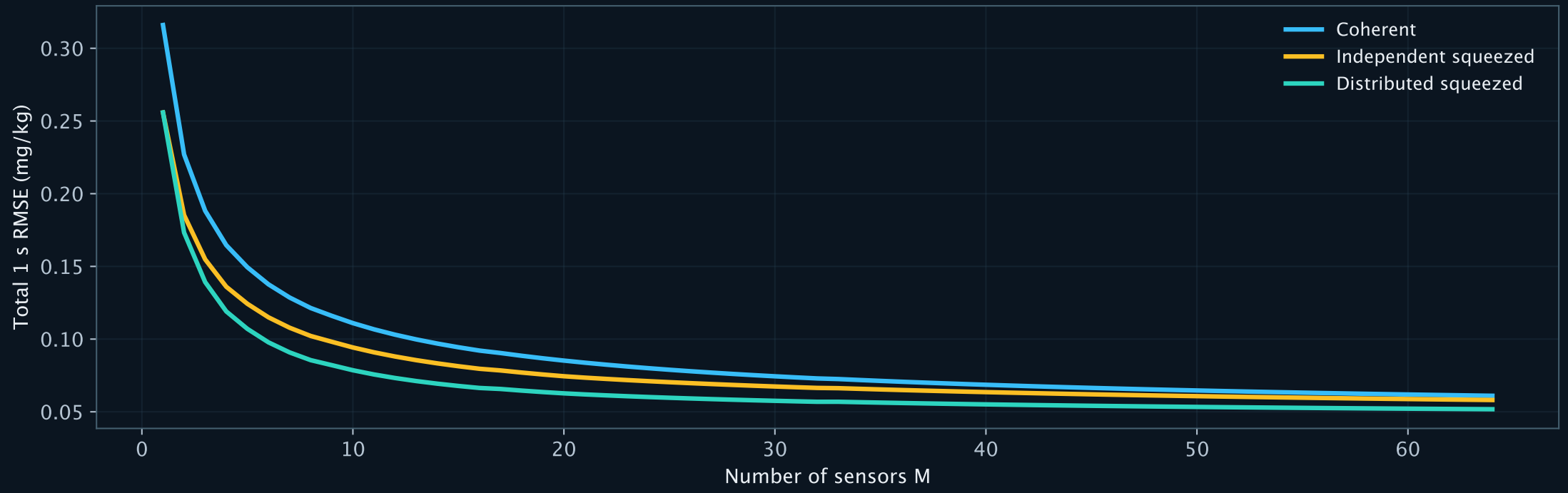


Fitted exponent, M=33–64: classical −0.50; distributed −0.95.

This theoretical reference exceeds the practical 10 dB source cap.

Loss and drift soften the scaling advantage

100 m links, 10 dB source cap, 10 mrad phase jitter and 0.05 mK shared residual



More sensors add probe power. The quantum source reaches its cap at 21 nodes.

Curves describe differential precision under the specified engineering assumptions.

The 16-node pilot reduces total error by 28%

One-second estimates, same 160 nW total launched probe power

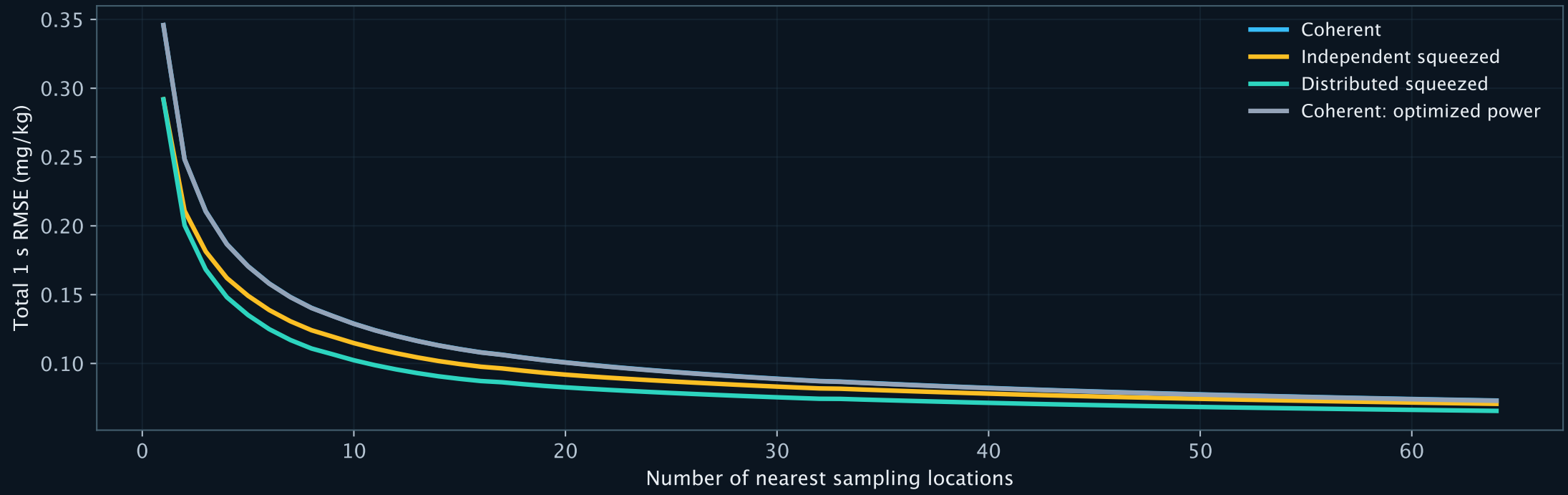


Total RMSE: 0.092 → 0.066 mg/kg; 1.39× precision gain.

Independent squeezers reach 0.080 mg/kg: the distributed configuration adds a further 17% reduction.

Real distances reduce the regional gain

64 distinct supplied well locations; an assumed hub and 1.3× routing overhead



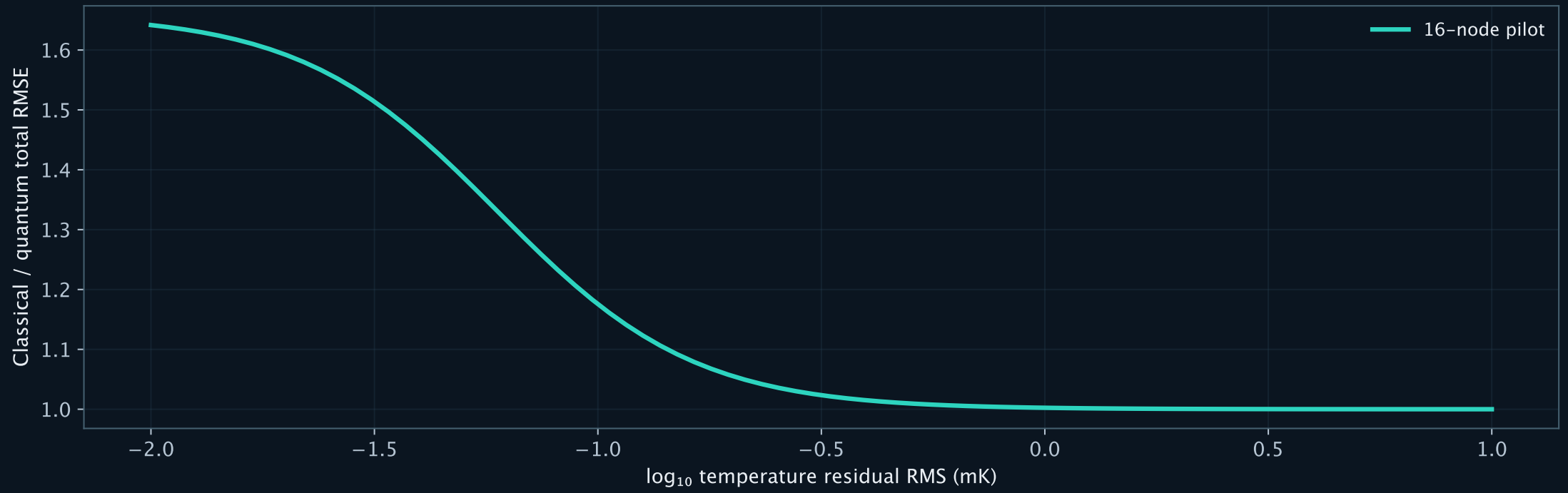
Locations span 3.5–16.6 km from the hub; mean transmission at M=64 is 37%.

At 64 nodes: 1.11× gain over optimized coherent power allocation.

The supplied file is USGS-labelled; missing provenance prevents independent field-data validation.

Temperature compensation determines useful gain

Common thermal error appears in every sensor and survives spatial averaging

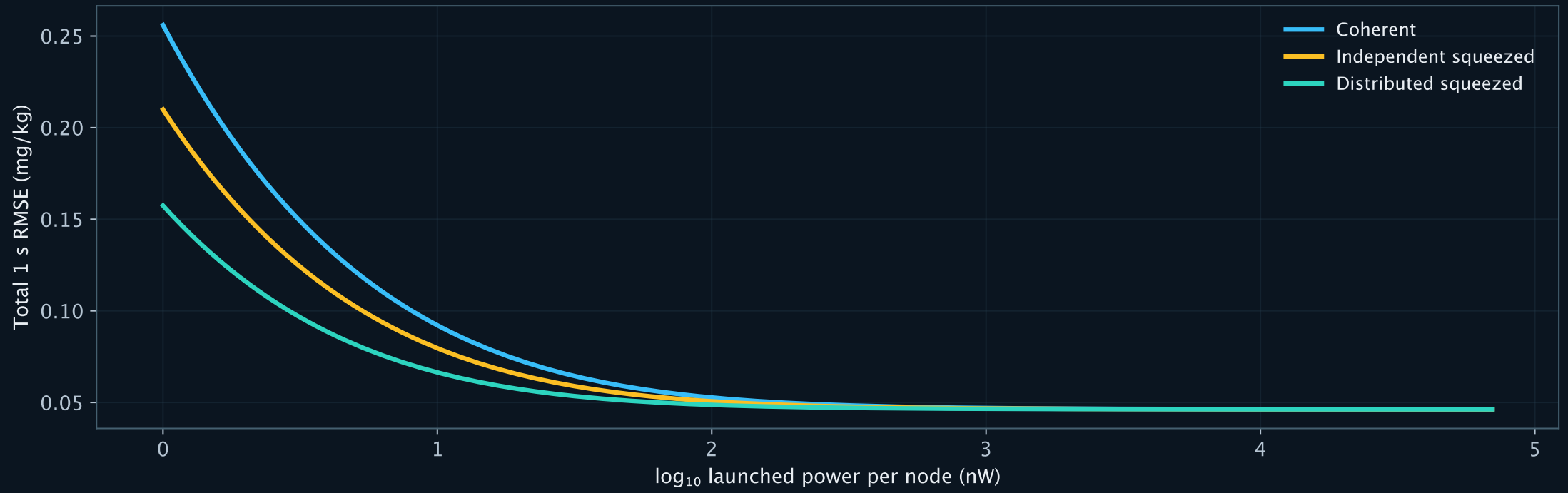


1 mK corresponds to about 0.916 mg/kg of apparent salt change.

At a 10 mK common residual, quantum and classical total errors are almost identical.

More classical light is a competing option

Power sweep from 1 nW to 70 µW per node; no claim that water requires a 10 nW limit

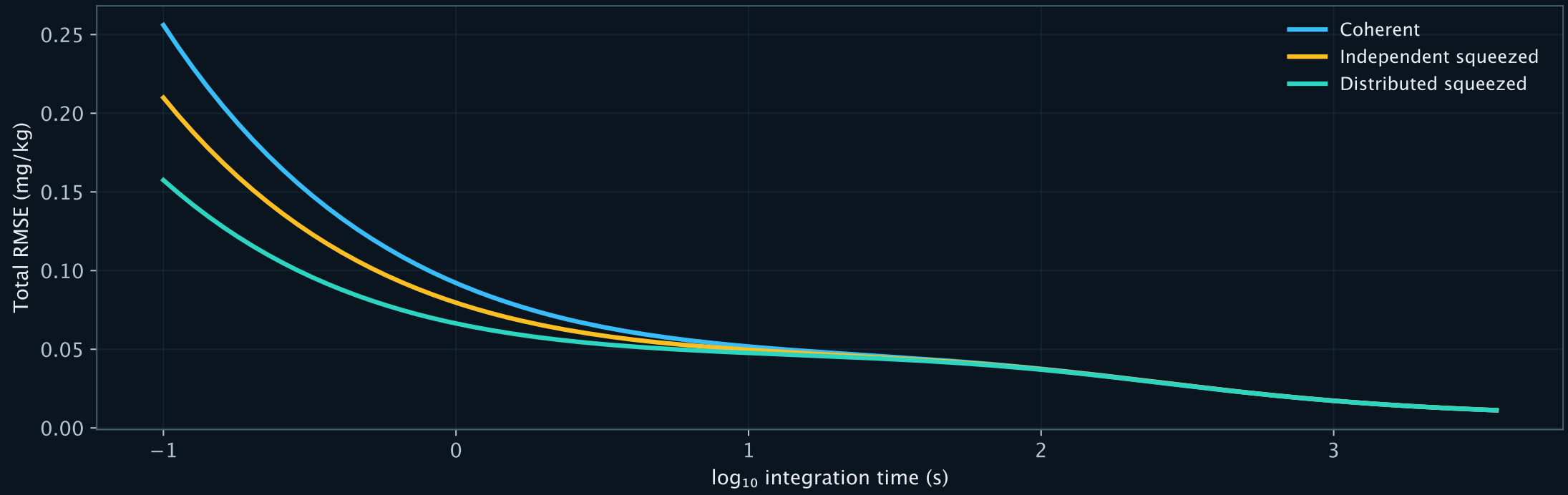


~28 nW/tap coherent matches the 10 nW/tap distributed total RMSE.

Same transducer and drift assumptions. The low-dose requirement needs justification.

Continuous averaging encounters slow drift

Exact variance of a rectangular average of a 60-second correlated temperature process

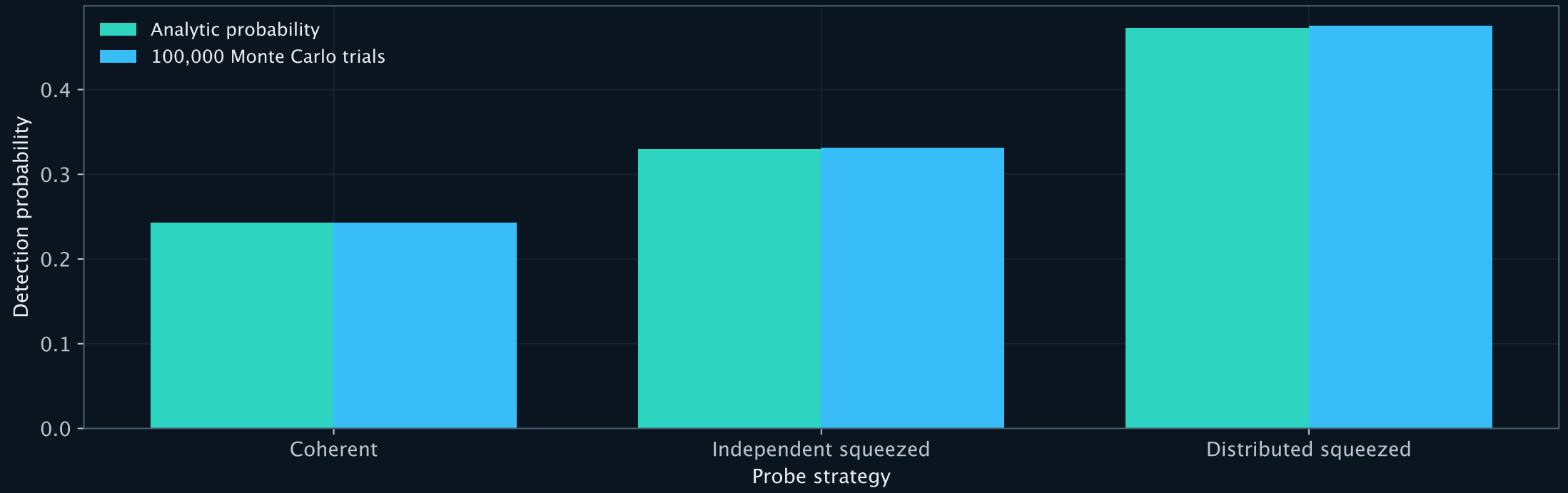


Short windows are optical-noise limited; longer windows encounter correlated drift.

Fixed calibration offsets average across nodes, not across time. Long windows also blur fast events.

Quantum readout improves a weak-change decision

A 0.15 mg/kg regional increase; identical 1% false-alarm probability per decision

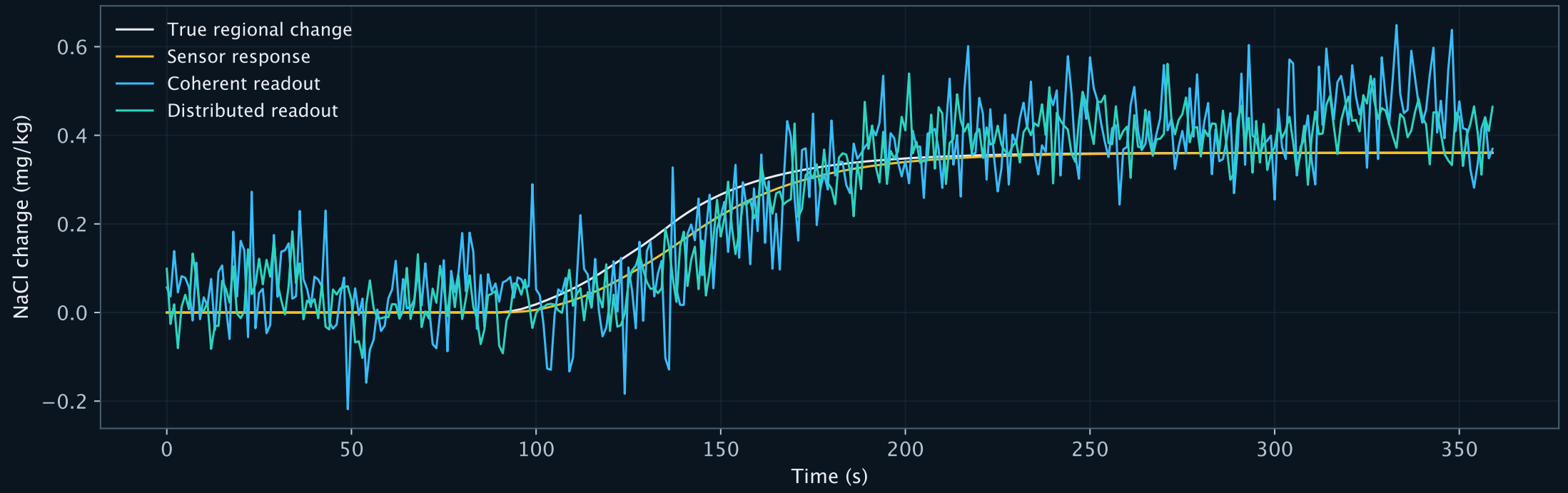


Detection rises from 24% to 47%; the same signal is easier to distinguish.

1% at 1 Hz means 36 expected flagged windows/hour. An operational alarm needs sequential validation.

Continuous monitoring of unequal local changes

Synthetic arrivals differ across all 16 taps; the regional estimator remains well defined

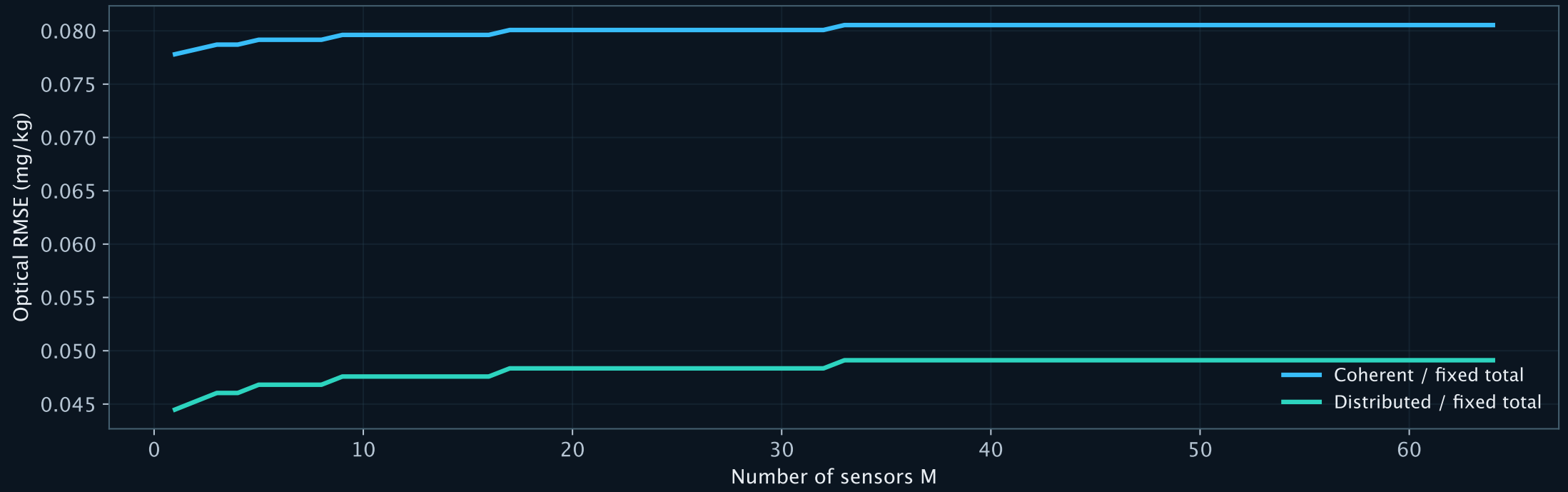


The shared temperature trace and calibration realization are matched between strategies.

A 10-second response introduces lag. Stationary precision does not include this transient bias.

Sensor count and photon budget are distinct

Fixed 160 nW total carrier-plus-squeezing budget and fixed total squeezing resource



Adding nodes spreads the same light more thinly; it does not produce a free 1/M improvement.

Sensor locations set coverage. Fewer sensors cannot guarantee the same protection against unsampled hotspots.

A real optical experiment anchors the opportunity

Reported refractive-index sensitivities; a separate measured benchmark



795 nm, air, 199 kHz ultrasound. *Best single-beam classical value is inferred in the paper.

Twin beams give 1.24× sensitivity gain over that best classical comparator; not a NaCl field test.

A credible next experiment

Validate a four-tap optical prototype before extrapolating to a regional network

Calibrate NaCl phase response with standards and controlled temperature at 1550 nm.

Measure noise spectra, sample loss, modulation efficiency and squeezing at the same optical dose.

Compare coherent, independent squeezed and distributed readout using blind injected changes.

Validate drift, fouling and false alarms over hours; measure actual optical and electrical resources.

Advance only if total error improves. Keep chemical identification and spatial coverage as separate requirements.

Assumptions that control the result

These are design targets, not measured specifications of a quantum water detector

10 nW/node; 10 μm effective optical interaction; 500 mg/kg background; 1-second decisions.

10 dB maximum source squeezing; 1 MHz effective mode rate; 10 mrad phase-lock RMS.

95% detector, 90% sample, 90% coupling; 0.05 dB splitter excess/stage.

100 m pilot links; regional links = 1.3× geographic distance; 0.18 dB/km fiber anchor.

0.05 mK compensated common residual; 60 s correlation; 0.03 mg/kg independent calibration RMS.

10 mK field stress test; 1 nW–70 μW power sweep. Full register: docs/SOURCES.md.

Sources and reproducibility

Code, parameter provenance, raw numerical outputs and figures are included

Dowran et al. (2018), Optica 5, 628 — doi:10.1364/OPTICA.5.000628

Zhuang et al. (2018), PRA 97, 032329 — arXiv:1711.10459

Oh et al. (2020), PR Research 2, 023030 — doi:10.1103/PhysRevResearch.2.023030

Guo et al. (2020), Nature Physics 16, 281 — arXiv:1905.09408

Saunders et al. (2016), Applied Optics 55, 947 — doi:10.1364/AO.55.000947

Mehmet et al. (2011), Optics Express 19, 25763 — arXiv:1110.3737

Kim et al. (2012), Optical Sensors STu1F.3; Corning AN2008; YSI EXO specifications.

Water QML paper (2026), Quantum Reports 8, 60 — doi:10.3390/quantum8030060

How the simulation runs

Classical CPU computation of a proposed quantum optical sensor

Inputs: default.json assumptions and a CSV of 64 distinct coordinates.

Python 3.12: NumPy covariance, SciPy probability, pandas data, Matplotlib plots.

Compute phase sensitivity and loss, then mean variance plus drift and calibration.

Outputs: RMSE, detection probabilities, synthetic time traces and ten figures.

Development and validation

Published physics, explicit assumptions and reproducible comparisons

model.py implements the equations; run.py sweeps count, loss, power and drift.

Analytic RMSE curves; 100,000 null/event draws check Gaussian detection.

13 checks test physicality, limits, units and agreement; seed 2601006.

Run bash src/reproduce.sh. No quantum device or trained model was executed.